MODULE I

ORDINARY DIFFERENTIAL EQUATIONS OF THE FIRST AND SECOND ORDERS

Module Introduction

This module deals with ordinary differential equations. In the first unit we will look at the ordinary differential equations of the first order. In the second unit we will deal with the ordinary linear differential equations of the second order. By doing so students will be able to express terms and concepts related to ordinary differential equations and get ready to attend Module II.

Module Objectives:

At the end of this module students will be able to:

- Understand the concept of ordinary differential equations.
- Define ordinary differential equations.
- Distinguish linear and non-linear, homogeneous and non homogeneous, separable and exact differential equations and demonstrate together with solving exercises.
- Compute general and particular solutions of linear differential equations
- Demonstrate the application of differential equations together with solving exercise
- Apply the concept of differential equations in solving real life problems.
- Appreciate the applications of differential equation

UNIT ONE

ORDINARY DIFFERENTIAL EQUATIONS OF THE FIRST ORDER

Unit Introduction

This unit is divided into five sections. The first section presents *basic concepts and ideas of ordinary differential equations*. The second section discusses *separable differential equations and reduction to separable form. Exact differential equations and integrating factors* are parts of the third section. *Linear first order differential equations and reduction to linear form (Bernoulli Equation)* will be treated in the fourth section and *Some Special Differential Equations* are parts of the last section.

Unit Objectives:

At the end of the unit students will be able to:

- Define ordinary differential equations of the firs order.
- Distinguish separable and exact, differential equations
- Identify linear and non-linear differential equations
- Demonstrate how to differentiate separable and non separable, exact and non exact, linear and non linear first order differential equations together with solving exercises
- Apply the concepts of ordinary linear differential equations of the firs order in solving real life problems

1.1. Basic Concepts and Ideas of Ordinary Differential Equations

Definition:

Ordinary differential equation (ODE) is an equation which involves one or several derivatives of an unspecified function y of x $(y = f(x))$; the equation may also involve y itself and constants.

For example,

a) $y' = \cos x$

b) $y' = -32x$

c) $y'' + 4y = 0$

d) $y'' + \sin y = 0$

e) $y' + 5y = 12e^{7x}$

f) $y^2 y''' y' + 2e^x y'' = (x^2 + 3)y^2$,

are ordinary differential equations.

Remarks:

1. A differential equation is said to be of order n if the nth derivative of y is the highest derivative in the equation.
   For instance,
   a) $y'' + 4y' + by = 0$ is the second order differential equation.
   b) $y''' + y''' + by = e^x$ is the third ordered differential equation

2. A function $y = g(x)$ is a solution of a given first order differential equation if it becomes an identity when y and y' are replaced by g and g' respectively.

For example, the function $y = x^2$ is a solution of $xy' = 2y$ for all x, since $y' = g'(x) = 2x$, then substituting it we obtain that

$xy' = x(2x) = 2x^2 = 2y$.

3. Some times a solution of a differential equation will be given implicitly in the form $G(x, y) = 0$, and is called an implicit solution. For instance,

$$x^2 + y^2 - 1 = 0$$

is an implicit solution of the differential equation

$$yy' = -x \qquad (on \ -1 < x < 1.)$$

4. A first order differential equation may have more than one solution which can be represented by a single formula involving an arbitrary constant such a solution is called a *general solution*. If we assign a definite value to that constant, the solution is called a *particular* solution.

For example, each of the functions

$$y = \sin x, \qquad y = \sin x + 3, \qquad y = \sin x - \frac{4}{5}$$

is a particular solution of the differential equation

$$y' = \cos x.$$

Hence

$$y = \sin x + c$$

is a general solution, where c is a constant.

5. A first order differential equation together with an initial condition, $y(x_0) = y_0$, is called the initial value problem.

For example,

$$y^2xy' - x + 1 = 0, \quad y(1) = 3$$

is an initial value problem. ◂

Activity 1.1

1. Find the curve through the point $(1,1)$ in the xy-plane having at each of its points the slope $-\frac{y}{x}$.

2. Solve the following differential equations.

   a) $y' = x^3,$ b) $y'' = \sin 2x,$ c) $y' = e^{-\frac{x}{2}},$

3. Find a first order differential equation involving both y and y' for which the given functions is a solution

   a) $y = e^{-x^2}$ b) $y = x^4$

1.2 Separable Differential Equations; Reduction to Separable Form

1.2.1 Separable Differential Equations

Definition:

A differential equation of the form

$$g(y)y' = f(x)$$

or

$$g(y)dy = f(x)dx \qquad (1)$$

is called a separable differential equation.

Equation (1) is called separable because the variables x and y are separated.

Note: By integrating both sides of the above equation one can easily obtain the general solution.

Example: Solve the following differential equations.

a) $y' = 1 + y^2$ b) $xy' + y = 0, \quad y(1) = 1$

Solutions:

a) $$y' = 1 + y^2 \quad \Rightarrow \quad \frac{dy}{dx} = 1 + y^2$$

$$\Rightarrow \quad \frac{dy}{1 + y^2} = dx$$

Integrating both sides, we obtain

$$\arctan y = x + c \quad \Rightarrow \quad y = \tan(x + c). \qquad \blacktriangleleft$$

b) $xy' + y = 0, \quad y(1) = 1$

$\Rightarrow xy' = -y$

$\Rightarrow x\dfrac{dy}{dx} = -y$

$\Rightarrow \dfrac{dy}{y} = \dfrac{dx}{x}.$

Integrating both sides we obtain

$$\ln|y| = -\ln|x| + c.$$

From this we obtain the general solution

$$y = \frac{c}{x}.$$

By using the initial condition, $y(1) = 1,$ we see that $c = 1$ and hence

$$y(x) = \frac{1}{x}.$$ ◂

Activity: 1.2.1

1. Solve the following differential equations.

a) $y' = e^x y^3$

b) $yy' = \sin^2 x, \quad y(0) = \sqrt{3}$

c) $y' = \pi(1 + y^2)$

d) $y' = y\tanh x. \quad y(0) = \dfrac{7}{5}$

e. $L\left(\dfrac{dI}{dt}\right) + RI = 0,\ I(0) = I_0,$ *where L and R are constants*

f) $y^2 xy' - x + 1 = 0, \quad y(1) = 3$

g) $(x\ln x)y' = y$

h) $y' = -\dfrac{x}{y}, \quad y(2) = \sqrt{5}$

i) $(x^2 + 1)y + y^2 + 1 = 0, \quad y(0) = 1$

2. (*Newton's Law of Cooling*). A copper ball is heated to a temperature of $100^0 c$. Then at a time $t = 0$ it is placed in water which is maintained at a temperature of $30^0 c$. At the end of 3 minutes the temperature of the ball is reduced to $70^0 c$. Find the time at which the temperature of the ball is reduced to $31^0 c$. (*Hint: the rate of change of the temperature T of the ball is proportional to the difference between T and the temperature of the surrounding medium.*)

1.2.2 Reduction to Separable Form

Certain first order differential equations are not separable but can be made separable by a simple change of variables. This holds for equations of the form:

$$y' = g\left(\frac{y}{x}\right). \qquad (*)$$

where g is any given function of $\frac{y}{x}$ .To solve such equations we set $u = \frac{y}{x}$, assuming that y and u are functions of x. Then $y = ux$ and hence

$$y' = u'x + u\,.$$

Substituting this in $(*)$ we obtain

$$u'x + u = g(u),$$

thus

$$\frac{du}{dx}x = g(u) - u$$

This can be separated as

$$\frac{du}{g(u) - u} = \frac{dx}{x}.$$

If we integrate this and replace u by $\frac{y}{x}$, we obtain the general solution of $(*)$.

Example: Solve the differential equation

$$2xyy' - y^2 + x^2 = 0.$$

Solution: Since

$$2xyy' - y^2 + x^2 = 0$$

Can be re written as

$$2xyy' = y^2 - x^2$$

$$\Rightarrow 2y' = \frac{y}{x} - \frac{x}{y}. \qquad (*)$$

Now let $u = \frac{y}{x}$.

$$\Rightarrow y = ux$$

$$\Rightarrow y' = u'x + u.$$

Then from $(*)$ it follows that

$$2(u'x+u) = u - \frac{1}{u}$$

$$\Rightarrow \frac{2du}{dx}x = -\frac{u^2+1}{u}$$

$$\Rightarrow \frac{2u\,du}{u^2+1} = -\frac{dx}{x}$$

Integrating both sides yields

$$\ln|u^2+1| = \ln|x| + c$$

$$\Rightarrow u^2+1 = \frac{c}{x}$$

Now, setting $u = \frac{y}{x}$ we obtain

$$y^2 + x^2 = cx.$$

Method of Transformation of Variables

First order differential equations of the form

$$y' = f(ax+by+c), \qquad (*)$$

where a, b and c are real constants, can be made separable by substituting

$$z = ax+by+c.$$

From this, it follows that

$$z' = a+by'.$$

That is,

$$y' = \frac{z'-a}{b}.$$

Substituting this in (*) we get,

$$\frac{z'-a}{b} = f(z).$$

This can be separated as

$$\frac{dz}{a+bf(z)} = dx.$$

Example: Solve the following differential equations

a) $y' = (x+y+1)^2 - 2$ b) $y'+1 = \sqrt{x+y+2}$, $y(0) = 2$

Solution:

a) Letting

$$z = x+y+1,$$

we have

$$y' = z'-1.$$

Substituting this in the given equation, we get

$$z' - 1 = z^2 - 2.$$

This implies

$$\int \frac{dz}{z^2 - 1} = \int dx,$$

$$\Rightarrow \ln\left|\frac{z-1}{z+1}\right| = 2x + c$$

$$\Rightarrow \frac{z-1}{z+1} = ce^{2x}.$$

Hence

$$z = \frac{1 + ce^{2x}}{1 - ce^{2x}}.$$

But since

$$z = x + y + 1,$$

then

$$y = z - x - 1 = \frac{1 + ce^{2x}}{1 - ce^{2x}} - x - 1$$

$$\therefore\ y(x) = \frac{1 + ce^{2x}}{1 - ce^{2x}} - x - 1. \qquad \blacktriangleleft$$

b) $y' + 1 = \sqrt{x + y + 2},\quad y(0) = 2$

Let

$$z = x + y + 2 \qquad (*)$$

$$\Rightarrow y = z - x - 2$$

$$y' = z' - 1.$$

Substituting this in the given equation, we get

$$z' = \sqrt{z}\,.$$

This implies

$$\int \frac{dz}{\sqrt{z}} = \int dx,$$

$$\Rightarrow \quad 2\sqrt{z} = x + c\,.$$

Substituting this in (*) we get

$$\sqrt{x+y+2} = \frac{x+c}{2},$$

$$\Rightarrow x+y+2 = \left(\frac{x+c}{2}\right)^2.$$

Thus

$$y(x) = \left(\frac{x+c}{2}\right)^2 - x - 2\,.$$

But $y(0) = 2$ implies

$$c = 4$$

Thus

$$y(x) = \left(\frac{x+4}{2}\right)^2 - x - 2\,. \qquad \blacktriangleleft$$

Activity 1.2.2

1. Solve the following differential equations.

a) $y' = \frac{y}{x} + \frac{2x^3 \cos x^2}{y},\ y(\sqrt{\pi}) = 0.$

b) $xy' = y + x\sec\left(\frac{y}{x}\right)$

c) $(x^3 + 3xy^2)dx + (y^3 + 3x^2 y)dy = 0$

d) $xyy' = 2y^2 + 4x^2,\ y(2) = 4$

e) $xy' = y + x^2 \sec\left(\frac{y}{x}\right),\ y(1) = \pi.$

f) $y' = \frac{y - x}{y + x},\ y(1) = 1.$

g) $y' = \frac{x + y}{x}$

h) $y' + \sin^2(x + y) = 0$

i) $y' = \frac{y^2 + 2xy}{x^2}$

j) $y' = (x + y - 2)^2$

2. Find the curve that passes through the point $(\sqrt{e}, \sqrt{e})$ and has the slope $\frac{x}{y} + \frac{y}{x}$

3. Solve $y' = \frac{1 - 2y - 4x}{1 + y + 2x}$. (Hint: use $v = y + 2x$).

1.3. Exact Differential Equations; Reduction to Exact Form

1.3.1 Exact Differential Equations

Definition:

A first order differential equation of the form

$$M(x,y)dx+N(x,y)dy=0$$

is said to be exact if there exists a function $u(x,y)$ such that

$$\frac{\partial u}{\partial x} = M, \quad \frac{\partial u}{\partial y} = N.$$

Note**:** $$\frac{\partial u}{\partial x} = M \Rightarrow u(x, y) = \int M(x,y)dx + k(y)$$

or $$\frac{\partial u}{\partial y} = N \Rightarrow u(x, y) = \int N(x,y)dy + l(x)$$

Theorem:

The necessary and sufficient condition for the differential equation

$$M(x,y)dx+N(x,y)dy=0 \qquad (*)$$

to be exact is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Proof: Exercise.

Remarks:

1. Every separable differential equation is exact.

2. The solution of the exact differential equation $(*)$ is given by $u(x, y) = c$, where c is a constant.

Example: Solve the following differential equations.

$$2x\sin 3y\,dx + (3x^2\cos 3y + 2y)dy = 0$$

Solution: Since

$$M(x, y) = 2x\sin 3y, \qquad N(x, y) = 3x^2\cos 3y + 2y$$

and

$$\frac{\partial M}{\partial x} = 6x\cos 3y, \quad \frac{\partial N}{\partial x} = 6x\cos 3y$$

we get

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

which implies the differential equation is exact.

But since

$$u(x, y) = \int M(x, y)dx + k(y) = \int 2x\sin 3y dx + k(y)$$

$$= x^2\sin 3y + k(y) \qquad (*)$$

and

$$\frac{\partial u}{\partial y} = N$$

then $3x^2\cos 3y + k'(y) = 3x^2\cos 3y + 2y$

$$\Rightarrow\ k'(y) = 2y \ \Rightarrow\ k(y) = y^2 + c_1.$$

From $(*)$ it follows that

$$u(x, y) = x^2\sin 3y + y^2 + c_1 = c_2.$$

This implies

$$x^2 \sin 3y + y^2 = c,$$

where $c = c_2 - c_1$ is a constant.

Thus $x^2 \sin 3y + y^2 = c$

is a required solution. ◂

Activity 1.3.1

1. Find the exact differential equation if:

   a) $u = x^2 + 9y^2$ b) $u = e^{xy}$

2. Show that the following differential equations are exact and solve them.

   a) $\pi y + (\pi x + \arcsin y)y' = \sin x$

   b) $ye^x dx + (2y + e^x)dy = 0$

   c) $[(x+1)e^x - e^y]dx - xe^y dy = 0$

   d) $y^2 dx + 2xy dy = 0$

   e) $y' = -\dfrac{2xy^3 + 2}{3x^2 y^2 + 8e^{4y}}$

3. Solve the following initial value problems.

   a) $(y+3)dx + (x-2)dy = 0, \quad y(1) = -7$

   b) $(2xydx + dy)e^{x^2} = 0, \quad y(0) = 2$

   c) $Sinx \cosh y - y' \cos x \sinh y = 0, \quad y(0) = 0$

4. Find the curve having the given slope y' and passing through the given point (x_0, y_0).

   a) $y' = -\dfrac{y-2}{x-2}, \ (0,0)$ b) $y = -\dfrac{x+2}{y+1}, \ (-3,-1)$

1.3.2 Reduction to Exact Form and Integrating Factors

Definition:

If an equation of the form

$$M(x,y)dx+N(x,y)dy=0$$

is not exact, it can be made exact by multiplying by some function $F(x,y)\neq 0$. Such a multiplier (or a function) is called an *integrating factor*.

Rules for Finding Integrating Factors

Although an equation of the form $M(x,y)dx+N(x,y)dy=0$ has integrating factors, there is no general method of finding them. We now use the following Theorems (rules) for finding integrating factors.

Theorem 1:

When $\frac{1}{N}\left(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}\right)$ is a constant, say k, or a function of x alone, say $f(x)$, then an integrating factor, $F(x)$, will be $e^{\int k dx}$ or $e^{\int f(x)dx}$.

Proof. Exercise.

Example: Solve the differential equation

$$(4x+3y^2)dx+2xydy=0.$$

Solution: Since

$$M(x,\ y) = 4x+3y^2,\quad N(x,y)=2xy$$

we obtain

$$\frac{\partial M}{\partial y} = 6y,\quad \frac{\partial N}{\partial x} = 2y$$

$$\Rightarrow\quad \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}.$$

Thus the differential equation is not exact.

But since

$$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{1}{2xy}(by - 2y) = \frac{2}{x},$$

an integrating factor is

$$F(x) = e^{2\int \frac{dx}{x}} = x^2.$$

Thus

$$x^2\left((4x+3y^2)dx+2xydy\right)=0$$

$$=(4x^3 + 3x^2y^2)\,dx+2x^3ydy = 0$$

is an exact differential equation, since

$$\frac{\partial}{\partial y}\left(4x^3+3x^2y^2\right) = 6x^2y = \frac{\partial}{\partial x}\left(2x^3y\right).$$

Now then, since

$$u(x,y) = \int (4x^3 + 3x^2y^2)dx + k(y)$$

$$= x^4 + x^3y^2 + k(y) \qquad (*)$$

and

$$\frac{\partial u}{\partial y} = 2x^3y$$

we have

$$2x^3y + k'(y) = 2x^3y\,.$$

This implies

$$k'(y) = 0 \Rightarrow k(y) = c_1,$$

where c_1 is a constant. Thus from $(*)$ it follows that

$$u(x, y) = x^4 + x^3y^2 + c_1 = c_2$$

$$\Rightarrow\ x^4 + x^3y^2 = c,$$

where $c = c_2 - c_1$ is a constant.

Therefore,

$$x^4 + x^3y^2 = c\,,$$

is a required solution. ◂

Theorem 2:

When $\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)$ is a constant, say k, or a function of y alone, say $f(y)$, then an integrating factor, $F(y)$, will be $e^{\int k\,dy}$ or $e^{\int f(y)\,dy}$.

Proof: Exercise

Example: Solve the differential equation $2xydx + 3x^2dy = 0$.

Solution: Since

$$M(x, y) = 2xy\,, \qquad N(x, y) = 3x^2,$$

we have

$$\frac{\partial M}{\partial y} = 2x, \qquad \frac{\partial N}{\partial x} = 6x\,.$$

This implies

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Thus the differential equation is not exact.

But since

$$\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial N}{\partial x}\right) = \frac{1}{2xy}(6x - 2x) = \frac{2}{y},$$

an integrating factor is

$$F(y) = e^{2\int \frac{dy}{y}} = y^2.$$

Thus

$$y^2(2xydx + 3x^2dy) = 0$$

$$= 2xy^3dx + 3x^2y^2dy = 0$$

is an exact differential equation, since

$$\frac{\partial}{\partial y}(2xy^3) = 6xy^2 = \frac{\partial}{\partial x}(3x^2y^2).$$

Now then, since

$$u(x,y) = \int (2xy^3)dx + k(y)$$

$$= x^2y^3 + k(y) \qquad (*)$$

and

$$\frac{\partial u}{\partial y} = 3x^2y^2,$$

we obtain

$$3x^2y^2 + k'(y) = 3x^2y^2.$$

$$\Rightarrow \; k'(y) = 0 \;\Rightarrow\; k(y) = c_1,$$

where c_1 is a constant. Thus from $(*)$ it follows that

$$u(x,y) = x^2y^3 + c_1 = c_2$$

$$\Rightarrow x^2y^3 = c,$$

where $c = c_2 - c_1$ is a constant.

Therefore,

$$x^2y^3 = c,$$

is a required solution.

◂

Activity 1.3.2

1. Find integrating factors and solve the following differential equations.

   a) $(y-1)dx - xdy = 0$

   b) $y\cos x dx + 2\sin x dy = 0$

   c) $3x^2ydx + 2x^3dy = 0$

2. Find integrating factors and solve the following initial value Problems

   a) $ydx - xdy = 0, \qquad y(1) = 2$

   b) $3y^2dx + xdy = 0, \qquad y(1) = \frac{1}{2}$

3. Verify that y, xy^3 and x^2y^5 are integrating factors of $ydx + 2xdy = 0$ and solve.

1.4. Linear First Order Differential Equations; Reduction to Linear Form (Bernoulli Equation)

1.4.1 Linear First Order Differential Equations

Definition:

A first order differential equation is said to be linear if it can be written in the form

$$y' + f(x)y = r(x),$$

where f and r are continuous functions.

Theorem:

The general solution of the linear first order differential equation:

$$y' + f(x)y = r(x),$$

where f and r are continuous functions, is given by

$$y(x) = e^{-\int f(x)dx}\left[\int e^{\int f(x)dx} r(x)dx + c\right]$$

Proof: Let us write the equation

$$y' + f(x)y = r(x) \qquad (*)$$

in the form

$$\big(f(x)y - r(x)\big)dx + dy = 0 \qquad (**)$$

and, let

$$M(x,y) = f(x)y(x) - r(x), \quad N(x, y) = 1,$$

then $\frac{\partial M}{\partial y} = f(x), \qquad \frac{\partial N}{\partial x} = 0.$

This implies, for $f(x) \neq 0$,

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Thus, the differential equation (**) and hence (*) is not exact.

But since

$$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = f(x),$$

the integrating factor is

$$F(x) = e^{\int f(x)dx}.$$

Now, let

$$F(x) = e^{h(x)},$$

where $h(x) = \int f(x)dx.$ Multiplying (*) by this factor we obtain

$$e^{h(x)}(y' + f(x)y) = e^{h(x)}r(x).$$

Since $h'(x) = f(x)$, this may be writer as

$$\frac{d}{dx}\left(ye^{h}\right) = e^{h}r.$$

Integrating both sides we have

$$y(x)e^{h(x)} = \int e^{h(x)}r(x)dx + c,$$

that is,

$$y(x) = e^{-h(x)}\left(\int e^{h(x)}r(x)dx + c\right), \qquad h(x) = \int f(x)dx.$$

Or

$$y(x) = e^{-\int f(x)dx}\left(\int e^{\int f(x)dx}r(x)dx + c\right).$$

This is the general solution of (*). ◂

Example: Solve the differential equation:

$$xy' + y + 4 = 0.$$

Solution: Since

$$xy' + y + 4 = 0$$

can be rewritten as

$$y' + \frac{1}{x}y = \frac{-4}{x},$$

then

$$f(x) = \frac{1}{x}, \quad r(x) = \frac{-4}{x}.$$

Thus using the result of the above theorem, we see that

$$y(x) = e^{-\int \frac{dx}{x}} \left[\int e^{\int \frac{dx}{x}} \left(\frac{-4}{x} \right) dx + c \right]$$

$$= e^{-\ln|x|} \left[-4 \int e^{\ln|x|} \left(\frac{1}{x} \right) dx + c \right]$$

$$= \frac{1}{x} \left[-4 \int dx + c \right]$$

$$= -4 + \frac{c}{x}.$$

Therefore,

$$y(x) = -4 + \frac{c}{x}$$

is the general solution. ◂

Activity 1.4.1:

1. Solve the following differential equations.

a) $y' \tan x = y$

b) $y' + 3y = e^{2x} + 6$

c) $y' = 2\dfrac{y}{x} + x^2 e^x$

e) $y' + y\cos x = \cos x$

f) $xy' + 2y = 2e^{x^2}$

g) $y' + 4y = \cos x$

1. Solve the following initial value problems.

a) $y' + y\tan x = 2x\cos x, \qquad y(0) = -1$

c) $xy' = (1+x)y, \qquad y(2) = 6e^2$

d) $xy' + 4y = 8x^4, \qquad y(1) = 2$

e) $xy' + y = \sin x, \qquad y\left(\dfrac{\pi}{2}\right) = \dfrac{2}{\pi}$

2. Find the curve having the given slope y' and passing through the given point (x_0, y_0) if

a) $y' = \dfrac{x}{4-4y}, \quad (0,2)$

b) $y' = \dfrac{1-x}{1+y}, \quad (1,0)$

1.4.2 Reduction to Linear Form (Bernoulli Equation)

Definition:

The differential equation of the form

$$y' + p(x)y = g(x)y^a,$$

where a is any real number, is called the Bernoulli equation.

Remark: For $a = 0$ and $a = 1$ the equation is linear, and otherwise it is non linear. For $a \neq 0,1$ if we set

$$u(x) = \left[y(x)\right]^{1-a}$$

the equation will be reduced to the linear form

$$u' + (1-a)p(x)u = (1-a)g(x),$$

which will be easily solved.

Example: Solve the Bernoulli equation

$$y' + y = y^2.$$

Solution: Since $a = 2$, by setting

$$u = y^{1-2} = y^{-1}, \qquad (*)$$

we obtain

$$u' = -y^{-2}y' \quad .$$

Thus

$$(y' + y = y^2)y^{-2}.$$

From this it follows that

$$y'y^{-2} + y^{-1} = 1,$$

or

$$-y'y^{-2} - y^{-1} = -1,$$

by multiplying both sides by -1.

Replacing $u = y^{-1}$ we obtain a linear first order differential equation

$$u' - u = -1.$$

With solution

$$u(x) = e^{\int dx}\left[-\int e^{-\int dx}dx + c\right]$$

$$= 1 + ce^{x} \ .$$

Using $(*)$ we obtain

$$y(x) = \frac{1}{u(x)} \ = \ \frac{1}{1+ce^{x}} \ .$$

This is the required general solution. ◂

Activity 1.4.2

Solve the following *Bernoulli* equations

a) $y' + \frac{1}{x}y = xy^2$

b) $y' + xy = xy^{-1}$

c) $3y' + y = (1-2x)y^4$

d) $y' = x^3y^2 + xy$

1.5 Some Special Differential Equations

1.5.1 Clairaut's Equation

Definition:

A differential equation of the form

$$y = xy' + f(y') \qquad (1)$$

is called Clairaut's equation.

To solve the Clairaut's equation we differentiate both sides of (1) to obtain

$$y' = y' + xy'' + f'(y')y''.$$

Thus

$$(x + f'(y'))y'' = 0$$

$$\Rightarrow y'' = 0 \quad \text{or} \quad x + f'(y') = 0.$$

i. If $y'' = 0$, then $y' = c$, where c is constant, and substituting this value into (1) we obtain

$$y(x) = cx + f(c),$$

which is a collection of straight lines.

ii. If $x + f'(y') = 0$, then $x = -f'(y')$, and (1) may be written as

$$y = f(y') - y'f'(y').$$

So if we substitute $y' = t$, we obtain the parameterized curve

$$x = -f'(t), \qquad y = f(t) - tf'(t).$$

This curve is also a solution, called the singular solution to clairaut's equation. ◂

Example: Solve the following Clairaut's equation.

$$y = xy' - e^{y'}$$

Solution: Differentiation both sides we get,

$$y' = y' + xy'' - e^{y'} y''$$

Thus

$$(x - e^{y'})y'' = 0.$$

$$\Rightarrow \; y'' = 0 \quad \text{or} \quad x - e^{y'} = 0.$$

i. $y'' = 0 \;\Rightarrow\; y' = c.$

Then substituting this in the equation, we obtain the general solution

$$y(x) = cx - e^{c}.$$

ii. $x - e^{y'} = 0 \;\Rightarrow x = e^{y'}$

$$\Rightarrow y' = \ln|x|.$$

Then substituting this in the equation, we get

$$y(x) = x\ln|x| - x.$$

This is the singular solution and the parameterized curves are

$$x = e^{t} \quad , \qquad y = e^{t}(t-1). \qquad ◂$$

Activity 1.5.1

Solve the following C*lairaut's* equations

a) $y = xy' + \ln y'$

b) $y = xy' + (y')^3$

1.5.2 Riccati Equation

Definition:

A differential equation of the form

$$y' + y^2 + f(x)y + g(x) = 0 \qquad (1)$$

is called Riccati equation.

To solve Riccati equation, we need to change it into a linear second order differential equation, which can be solved easily in the second chapter.

Now let

$$y = \frac{z'}{z}$$

Then

$$y' = \frac{z''}{z} - \left(\frac{z'}{z}\right)^2 .$$

So (1) becomes

$$\frac{z''}{z} - \left(\frac{z'}{z}\right)^2 + \left(\frac{z'}{z}\right)^2 + f(x)\left(\frac{z'}{z}\right) + g(x) = 0 .$$

Multiplying this equation by z, we obtain the linear second order equation

$$z'' + f(x)z' + g(x)z = 0 \qquad (2)$$

If the general solution to (2) can be found (*to be discussed in Chapter 2*), the general solution to (1) will be obtained from quotient

$$y = \frac{z'}{z}. \qquad \blacktriangleleft$$

Example: Solve the following Riccati equation.

$$y' + y^2 - 1 = 0, \qquad y(0) = -\frac{1}{3}$$

Solution: Let

$$y = \frac{z'}{z}.$$

Then

$$y' = \frac{z''}{z} - \left(\frac{z'}{z}\right)^2.$$

Substituting in the equation and simplifying we get

$$z'' - z = 0.$$

Its general solution is

$$z(x) = c_1 e^x + c_2 e^{-x},$$

which is found by the method to be discussed in chapter 2. Thus the general solution to the differential equation under consideration is

$$y(x) = \frac{z'}{z} = \frac{c_1 e^x - c_2 e^{-x}}{c_1 e^x + c_2 e^{-x}}.$$

By using the initial condition $y(0) = -\frac{1}{3}$, we get $c_1 = 0$. Thus

$$y(x) = -1. \qquad \blacktriangleleft$$

Activity 1.5.2

Solve the following Riccati equations

a) $\frac{dx}{dt} + x^2 + 1 = 0$ b) $y' + y^2 - 2y + 1 = 0$ d) $y' + y^2 - y - 2 = 0$

c) $y' + y^2 + 3y + 2 = 0,\ y(0) = 1$ e) $y' + y^2 + 2y + 1 = 0,\ y(1) = 0$

1.5.3 Reduction of Second Order Differential Equations to First Order

A second order differential equation in which y is not explicitly present can some times be solved by putting

$$u = y' .$$

So that

$$y(x) = \int u(x)dx$$

can be obtained.

Example: Solve the differential equation:

$$xy'' = 2 + y'$$

Solution: Let

$$u = y' .$$

Then

$$u' = y'' .$$

Thus, substituting in the equation we get

$$xu' = 2 + u$$

$$\Rightarrow u' - \frac{1}{x}u = \frac{2}{x} .$$

This is a linear first order differential equation with solution

$$u(x) = e^{-\int \frac{dx}{x}} \left[\int e^{-\int \frac{dx}{x}} \left(\frac{2}{x} \right) dx + c \right]$$

$$= cx - 2 .$$

Thus

$$y(x) = \int u(x)dx = \int (cx - 2)dx \quad = \int (cx - 2)dx.$$

That is,

$$y(x) = c_1x^2 - 2x + c_2.$$ ◂

Activity 1.5.3

2. Solve the following differential equations by reducing to the first order differential equations.

a) $xy'' + 2y' = x$

b) $1 - y' = 4y''$

c) $y'' + (y')^2 = 0$

d) $y'' = 1 + (y')^2$

1.6 Unit Summary

- Ordinary differential equation is an equation which involves one or several derivatives of an unspecified function y of x $(y = f(x))$; the equation may also involve y itself and constants.

- A differential equation of the form $g(y)dy = f(x)dx$ is called separable.

- A first order differential equations of the form

$$M(x, y)dx + N(x, y)dy = 0$$

is said to be exact if there exists a function $u(x, y)$ such that

$$\frac{\partial u}{\partial x} = M, \quad \frac{\partial u}{\partial y} = N$$

- The necessary and sufficient condition for the differential equation

$$M(x, y)dx + N(x, y)dy = 0$$

to be exact is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

- If an equation of the form $M(x, y)dx + N(x, y)dy = 0$ is not exact, it can be made exact by multiplying by an **integrating factor.**

- If $\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \begin{cases} k, & or \\ f(x) \end{cases}$,

then an integrating factor is $F(x) = \begin{cases} e^{\int k\,dx}, or \\ e^{\int f(x)\,dx} \end{cases}$

- If $\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = \begin{cases} c, or \\ f(y) \end{cases}$,

  then an integrating factor is $F(y) = \begin{cases} e^{\int c\,dx}, or \\ e^{\int f(y)dy} \end{cases}$

- The general solution of the linear first order differential equation

  $$y' + f(x)y = r(x),$$

  where f and r are continuous functions, is given by

  $$y(x) = e^{-\int f(x)dx}\left[\int e^{\int f(x)dx} r(x)dx + c\right]$$

- The differential equation of the form $y' + p(x)y = g(x)y^a$, where a is any real number, is called the *Bernoulli equation.*

- A differential equation of the form

  $$y = xy' + f(y')$$

  is called Clairaut's equation.

- A differential equation of the form

  $$y' + y^2 + f(x)y + g(x) = 0$$

  is called Riccati equation.

1.7 Review Problems

1. Solve the following differential equations.

   a) $xy' - y = 3x^4 \cos^2\left(\frac{y}{x}\right).$ b) $xy' - y - x^2 \tan\left(\frac{y}{x}\right) = 0$

   c) $9y' + (x + y - 1)^2 = 0$

2. Find the exact differential equation if $u = \tan(y^2 - x^2)$

3. Show that the following differential equations are exact and solve them.

   a) $(\cot y + x^2)dx = x \csc^2 y\, dy$

   b) $\frac{2xy}{1+y^2} y' + \ln(1 + y^2) = 0, \quad y(2) = \sqrt{e-1}$

4. Solve the initial value problem:

   $$(e^y + ye^x)dx + (e^x + xe^y)dy = 0, \quad y(1) = 0$$

5. Find integrating factors and solve the following differential equations.

   a) $2x \tan y dx + \sec^2 y dy = 0$

   b) $2Siny dx + \cos y dy = 0, \quad y(0) = \frac{\pi}{2}$

6. Solve the following differential equations.

   a) $y' + y = 2\sin x$ b) $y' + y \sin x = e^{\cos x}$

   c) $y' - (1 + 3x^{-1})y = x + 2, \quad y(1) = e - 1$

7. Solve the *Bernoulli* equation: $3y' + y = (1 - 2x)y^4$

8. Solve the C*lairaut's* equation: $y = xy' - \sqrt{y'}$

UNIT TWO
ORDINARY LINEAR DIFFERENTIAL EQUATIONS OF THE SECOND ORDER

Unit Introduction

This unit is divided into six sections. The first section presented *linear differential equations of the second order. General solutions and basis* are parts of the second section. *Second order homogeneous linear differential equations with constant coefficients* are discussed in the third section. *Euler-Cauchy equation* is presented in the fourth section, where as *Non homogeneous linear differential equations* are treated in the fifth section and s*ystems of differential equations* are parts of the last section.

Unit Objectives:

At the end of this unit, students will be able to:

- Define ordinary differential equations of the second order.
- Identify homogeneous and non homogeneous second order differential equations.
- Demonstrate how to compute general and particular solutions of second order differential equations together with solving exercises.
- Demonstrate how to transform systems of differential equations to second order differential equations together with solving exercises.
- Apply the concept of second order differential equations in solving real life problems.

2.1. Linear Differential Equations of the Second Order

Definition:

A differential equation of the form

$$y'' + f(x)y' + g(x)y = r(x), \qquad (1)$$

where f, g and r are continuous functions, is called a linear differential equation of the second order.

If $r(x) = 0$, then equation (1) becomes

$$y'' + f(x)y' + g(x)y = 0 \qquad (2)$$

and is said to be homogeneous differential equation.

If $r(x) \neq 0$, then equation (1) is said to be non homogeneous differential equation.

For example,

$$y'' + 4y = e^{-x} \sin x$$

is a non homogeneous linear differential equation, where as

$$(1-x^2)y'' - 2xy' + 6y = 0 \quad \text{and} \quad y'' + 4y = 0$$

are homogeneous linear differential equations.

Remark: Any differential equation of the second order which cannot be written in the form (1) is said to be nonlinear.

For example,

$$y''y + y' = 0 \quad \text{and} \quad y'' + \sqrt{y} = 0$$

are non linear differential equations .

Theorem1: (Fundamental Theorem for the Homogeneous Linear Differential Equation (2))

For *homogeneous linear* differential equation (2),any linear combination of two solutions on an open interval I is again a solution of (2) on I. In particular, for such an equation, sums and constant multiples of solutions are again solutions

Proof: Let y_1 and y_2 be solutions of (2) on I. Then by substituting

$$y = c_1 y_1 + c_2 y_2$$

and its derivatives,

$$y' = c_1 y_1' + c_2 y_2', \quad y'' = c_2 y_2'' + c_2 y_2''$$

into (2), we get

$$\begin{aligned} y'' + f(x)y' + g(x)y &= (c_1 y_1 + c_2 y_2)'' + f(x)(c_1 y_1 + c_2 y_2)' + g(x)(c_1 y_1 + c_2 y_2) \\ &= c_1 y_1'' + c_2 y_2'' + f(x)(c_1 y_1' + c_2 y_2') + g(x)(c_1 y_1 + c_2 y_2) \\ &= c_1(y_1'' + f(x)y_1' + g(x)y_1) + c_2(y_2'' + f(x)y_2' + g(x)y_2) \\ &= 0, \end{aligned}$$

since by assumption y_1, y_2 are solutions of (2). This shows that

$$y = c_1 y_1 + c_2 y_2$$

is a solution of (2) on I. ◂

Remark: The above theorem doesn't hold for *non homogeneous* linear differential equations or for *non linear* equations.

For example, substitution shows that:

1. The functions $y_1 = 1 + \cos x$ and $y_2 = 1 + \sin x$ are solutions of the non homogeneous linear differential equation $y'' + y = 1$, but $2(1 + \cos x)$, and $(1 + \cos x) + (1 + \sin x)$, are not solutions of this differential equation.

2. $y_1 = x^2$ and $y_2 = 1$ are solutions of the nonlinear differential equation. $y''y - xy' = 0$, but $-x^2$ and $x^2 + 1$ are not solutions of this differential equation.

Theorem2: (Existence and Uniqueness Theorem for Linear Initial Value Problems)

Let f, g and r be continuous functions on some open interval I, and let k_0, k_1 be two given constants. Then there exists a unique function $y(x)$ that satisfies the linear differential equation

$$y'' + f(x)y' + g(x)y = r(x) \tag{3}$$

on I and the two initial conditions

$$y(x_0) = k_0, \quad y'(x_0) = k_1 \tag{4}$$

For some value x_0 in I.

Proof. Exercise

Remark1: The conditions given in (4) both involve evaluations of the unknown function y and its derivatives at the same point x_0. This is a crucial requirement for the existence and uniqueness of a solution.

There is another type of problem involving the differential equations (3) in which conditions at more than one point are given. For example, we might specify $y(x_1)=d_1$ and $y'(x_2)=d_2$. Conditions of this sort are called boundary conditions, and a differential equation together with a set of boundary conditions is called a boundary value problem.

It is important to note that the existence and uniqueness of a solution guaranteed by Theorem1 for initial value problems, doesn't hold for boundary value problems.

Example: Observe that the boundary value problem

$$y''+y=0, \quad y(0)=y(\pi)=0,$$

has infinitely many solutions:

$$y = c\sin x, \quad \text{for any constant } c.$$

To check, note that $\sin 0 = \sin \pi = 0,$ so the boundary conditions are satisfied, and

$$(c\sin x)'' + c\sin x = -c\sin x + c\sin x = 0.$$

Where as the boundary value problem

$$y'' + y = 0, \quad y(0)=0, \quad y(\pi)=1$$

has no solution. On the other hand, the initial value problem

$$y'' + y = 0, \quad y(0)=0, \quad y'(0)=1$$

has the unique solution $y=\sin x$; the initial conditions hold since

$\sin 0=0$, $(\sin x)' = \cos x$, *and* $\cos 0 = 1$,and $(\sin x)'' + \sin x = -\sin x+\sin x=0.$

Thus if f, g and r are continuous functions, then the equation

$$y''+f(x)y'+g(x)y=r(x)$$

has a unique solution that satisfies the conditions

$$y(x_0)=k_0, \quad y'(x_0)=k_1$$

for any real number x_0, k_0 and k_1.

Remark 2: The requirement that the functions $f(x)$, $g(x)$ and $r(x)$ in (3) be continuous is also an essential requirement for the existence of a unique solution, as the following example demonstrates.

Example: Substitution shows that the function

$$y=cx^3+x$$

is a solution (for any constant c) of the initial value problem

$$x^2y''-3xy'+3y=0, \quad y(0)=0, \qquad y'(0)=1,$$

that is, the problem does not have a unique solution, it has infinitely many solutions. Theorem1 does not apply here, because for the differential equation

$$x^2y''-3xy'+3y=0, \quad y(0)=0, \qquad y'(0)=1,$$

that is, for the differential equation

$$y''-\frac{3}{x}y'+\frac{3}{x^2}y=0, \quad y(0)=0, \qquad y'(0)=1,$$

$$f(x)=\frac{-3}{x} \quad \text{and} \quad g(x)=\frac{3}{x^2}$$

are not continuous in an interval containing $x_0=0$.

Note that an *initial value problem* for a differential equation of the second order consists of the equation and two *initial conditions*

$$y(x_0)=k_1, \quad y'(x_0)=k_2.$$

2.2 General Solution and Basis

Definition:

A solution of a differential equation

$$y'' + f(x)y' + g(x)y = 0 \qquad (1)$$

on some open interval I is called a *general solution* if it is expressed in the form

$$y(x) = c_1 y_1(x) + c_2 y_2(x),$$

where c_1, and c_2 are constants and y_1 and y_2 form a basis (or a fundamental system) of solutions of equation (1) on I.

Definition:

Two *linearly independent* solutions of equation (1) on I are called *basis* of solutions on I.

Remark: y_1 and y_2 are linearly independent (or constitute such a fundamental system) on an interval I if and only if their quotient $\frac{y_1}{y_2}\left(or\ \frac{y_2}{y_1}\right)$ is not a constant on I, but depends on x.

Example:

1. Since $y_1 = e^x$ and $y_2 = 3e^x$ are linearly dependent, they don't form basis and hence there is no general solution.

2. Find a general solution of the differential equation:

$$y'' + 5y' + 6y = 0$$

Solution (2): The student may verify that $y_1 = e^{-2x}$ and $y_2 = e^{-3x}$ are solutions of the differential equation. Now since $y_1 = e^{-2x}$ and $y_2 = e^{-3x}$ are linearly independent, $\frac{y_1}{y_2} = e^{-5x} \neq consant$, they form a basis of solutions of the equation for all x, and a general solution is

$$y(x) = c_1 y_1(x) + c_2 y_2(x) = c_1 e^{-2x} + c_2 e^{-3x}$$ ◂

Theorem1:

Let $y_1(x)$ and $y_2(x)$ be linearly independent solutions of

$$y'' + f(x)y' + g(x)y = 0 \qquad (*)$$

and let $y_3(x)$ be another solution. Then there exist unique constant c_1 and c_2 such that

$$y_3(x) = c_1 y_1(x) + c_2 y_2(x) .$$

In other words, any solution of $(*)$ can be written as a linear combination of two given linearly independent solutions of $(*)$.

Proof: Exercise

How to Obtain a Basis if One Solution is Known (Reduction of Order)

Theorem2:

If one solution, y_1 of the differential equation

$$y'' + f(x)y' + g(x)y = 0 \qquad (*)$$

is known on some interval I, the second linearly independent solution y_2 of the equation can be obtain by the method of reduction of order. So that y_1, y_2 are the basis.

Proof: Suppose y_1 is a known nonzero solution of $(*)$, to get y_2, since y_1 and y_2 are linearly independent, $y_2 \neq ky_1$ for any constant k. So

$$\frac{y_2}{y_1} = u(x)$$

must be a non constant function of x, and $y_2 = uy_1$ must satisfy $(*)$. Now, substituting

$$y_2 = u\, y_1,$$

and its derivatives,

$$y_2' = u'y_1 + uy_1' \text{ and } y_2'' = u''y_1 + 2u'y_1' + uy_1''$$

into (*) gives

$$u''y_1 + 2u'y_1' + uy_1'' + f(x)(u'y_1 + uy_1') + g(x)\, uy_1 = 0.$$

Collecting terms in u'', u' and u, we have

$$u''y_1 + u'(2y_1' + f\, y_1) + u(y_1'' + f\, y_1' + g\, y_1) = 0$$

But since y_1 is a solution of $(*)$, the expression in the last parentheses is zero.

Thus

$$u''y_1 + u'(2y_1' + f\,y_1) = 0\,.$$

Dividing by $u'y_1$, we can rewrite this equation in the form

$$\frac{u''}{u'} = -2\frac{y_1'}{y_1} - f\,.$$

Setting $z = u'$, $z' = u''$, we obtain

$$\frac{z}{z} = -2\frac{y_1'}{y_1} - f$$

$$\Rightarrow \int\frac{dz}{z} = -2\int\frac{dy_1}{y_1} - \int f(x)dx$$

$$\Rightarrow \ln|z| = -2\ln|y_1| - \int f(x)dx\,.$$

Exponentiating and simplifying, we obtain

$$z(x) = \frac{1}{y_1^2(x)}e^{-\int f(x)dx}$$

But since $z = u'$, we have

$$u'(x) = \frac{1}{y_1^2(x)}e^{-\int f(x)dx}\,.$$

That is,

$$u(x) = \int\frac{e^{-\int f(x)dx}}{y_1^2(x)}dx\,.$$

Hence the desired solution is

$$y_2 = uy_1 = y_1(x)\int\frac{e^{-\int f(x)dx}}{y_1^2(x)}dx$$ ◀

Example: Find general solutions for the differential equation

$$x^2y'' - xy' + y = 0, \quad x > 0,$$

if one solution is $y_1 = x$.

Solution: Since $x^2y'' - xy' + y = 0$ Can be written as

$$y'' - \frac{1}{x}y' + \frac{1}{x^2}y = 0,$$

and $y_1 = x$ then

$$y_2 = x\int \frac{e^{-\int \frac{1}{x}dx}}{x^2}dx = x\ln x, \qquad \text{for } x > 0.$$

Further more, since

$$y_1 = x, \quad y_2 = x\ln x, \quad x > 0$$

are linearly independent, they are the basis for the given differential equation and the general solution is

$$y(x) = c_1x + c_2x\ln x, \qquad x > 0$$ ◂

Definition (Wronkian):

Let $y_1(x)$ and $y_2(x)$ be any two solutions to the differential equation

$$y'' + f(x)y' + g(x)y = 0.$$

The Wronkian of y_1 and y_1 is defined as

$$W(y_1, y_2)(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = y_1(x)y_2'(x) - y_1'(x)y_2(x).$$

Theorem3. (Wronkian Test**)**

Let y_1 and y_2 be solutions of

$$y'' + f(x)y' + g(x)y = 0$$

for x in an open interval I. Let f and g be continues on I. Then

1. Either $W(y_1, y_2)(x) = 0$ for all x in I or $W(y_1, y_2)(x) \neq 0$ for all x in I.
2. y_1 and y_2 are linearly independent on I if and only if $W(y_1, y_2)(x) \neq 0$ for all x in I.

Proof: Exercise.

Activity 2.2

1. Solve the following differential equations by the method of reduction of order if one solution, y_1 ,is given

   a) $x^2y'' - 5xy' + 9y = 0, \qquad y_1 = x^3$

   b) $y'' - \frac{3}{x}y' + \frac{4}{x^2}y = 0, \qquad y_1 = x^2$, for $x > 0$

   c) $y'' + 4y = 0, \quad y_1 = \cos(2x)$

   d) $(2x^2 + 1)y'' - 4xy' + 4y = 0, \qquad y_1 = x$, for $x > 0$

2. Find general solution of

$$y'' - \frac{2}{x}y' + \frac{2}{x^2}y = \frac{\ln x}{x}, \qquad x > 0$$

if the two solutions $y_1 = x$ and $y_1 = x^2$ for the corresponding homogeneous equations are given.

3. Application of reducible equations (Motion)

   a) A small body moves on straight line so that the product of its velocity and acceleration is constant, say, $1m^2/\sec^3$.If at $t = 0$ the body's distance from the origin is 2 meters and its velocity is $2m/\sec$, what are the distance and velocity at $t = 6\sec$?

   b) What happens in the above problem (prob.2a), if the acceleration equals the velocity (the other data being as before)? Will the distance at $t = 6$ be larger?

5. Find the curve through the origin in the xy-plan which satisfies $y'' = 2y'$ and whose tangent at the origin has slope 1.

2.3 Second Order Homogeneous Linear Differential Equations with Constant Coefficients

We shall now see how to solve the second order homogeneous linear differential equation of the form

$$y'' + by' + cy = 0, \qquad (1)$$

where coefficients b and c are constant. To solve (1) we remember from Unit One that a *first - order* linear differential equation $y' + ky = 0$ with constant coefficient k has an exponential function as a solution, $y = e^{-kx}$. This gives us the idea to try as a solution of (1) the function

$$y = e^{\lambda x}. \qquad (2)$$

Substituting (2) and its derivatives

$$y' = \lambda e^{\lambda x} \quad \text{and} \quad y'' = \lambda^2 e^{\lambda x}$$

into equation (1), we obtain

$$\lambda^2 + b\lambda + c = 0. \qquad (3)$$

Hence (2) is a solution of (1), if λ is a solution of the quadratic equation (3). The equation $\lambda^2 + b\lambda + c = 0$ is called the characteristic equation (or auxiliary equation) of (1). Its roots are

$$\lambda_1 = \frac{1}{2}\left(-b + \sqrt{b^2 - 4c}\right), \quad \lambda_2 = \frac{1}{2}\left(-b - \sqrt{b^2 - 4c}\right).$$

Our derivation shows that the functions

$$y_1 = e^{\lambda_1 x} \quad \text{and} \quad y_2 = e^{\lambda_2 x}$$

are solutions of (1).

Real Roots, Complex Roots and Double Root of the Characteristic Equation

Since the characteristic equation of $y'' + by' + cy = 0$ is $\lambda^2 + b\lambda + c = 0$, then

$$\lambda_1 = \frac{1}{2}\left(-b + \sqrt{b^2 - 4c}\right), \quad \lambda_2 = \frac{1}{2}\left(-b - \sqrt{b^2 - 4c}\right).$$

But since b and c are real constants, depending on the sign of the discriminant $b^2 - 4c$, the characteristic equation may have

Case I.	Two distinct real roots if $b^2 - 4c > 0$,
Case II	Two complex conjugate roots if $b^2 - 4c < 0$,
Case III	A real double root if $b^2 - 4c = 0$.

Case I: Two Real Roots

In this case, since $b^2 - 4c > 0$, then λ_1 and λ_2 are distinct real roots and hence

$$y_1 = e^{\lambda_1 x} \quad \text{and} \quad y_2 = e^{\lambda_2 x}$$

are the basis and hence the corresponding general solution is

$$y(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

where c_1 and c_2 are constants. ◀

Example: Solve the differential equation

$$y'' + y' - 2y = 0, \quad y(0) = 4, \quad y'(0) = 1$$

Solution: The characteristic equation is

$$\lambda^2 + \lambda - 2 = 0,$$

and hence its roots are

$$\lambda_1 = \frac{1}{2}\left(-1 + \sqrt{9}\right) = 1, \qquad \lambda_2 = \frac{1}{2}\left(-1 - \sqrt{9}\right) = -2$$

So that the general solution is

$$y(x) = c_1 e^{-2x} + c_2 e^{x}.$$

But since

$$y(0) = 4, \qquad y'(0) = 1$$

and

$$y'(x) = -2c_1 e^{-2x} + c_2 e^{x},$$

then

$$y(0) = c_1 + c_2 = 4, \qquad y'(0) = -2c_1 + c_2 = 1.$$

From these it follows that

$$c_1 = 1, \qquad c_2 = 3$$

Therefore, the particular solution is

$$y(x) = e^{-2x} + 3e^{x}$$ ◀

Case II: Two Complex Roots

In this case, since $b^2 - 4c < 0$, then the roots are complex conjugates,

$$\lambda_1 = p + iq, \quad \lambda_2 = p - iq,$$

where p and q are real constants and $q \neq 0$. But then since

$$s_1 = e^{(p+iq)x} = e^{px} e^{iqx} = e^{px} \cos qx + ie^{px} \sin qx,$$

and

$$s_2 = e^{(p-iq)x} = e^{px}\, e^{-qx} = e^{px}\cos qx - ie^{px}\sin qx$$

then,

$$y_1 = \frac{1}{2}(s_1 + s_2) = e^{px}\cos qx, \qquad y_2 = \frac{1}{2i}(s_1 - s_2) = e^{px}\sin qx.$$

which are linearly independent solutions and therefore are the basis. Hence the corresponding general solution is

$$y(x) = e^{px}(A\cos qx + B\sin qx),$$

where A and B are constants. ◂

Example: Solve the differential equation

$$y'' - 2y' + 10y = 0, \;\; y(0) = 4, \qquad y'(0) = 1$$

Solution: The characteristic equation is

$$\lambda^2 - 2\lambda + 10 = 0$$

and its roots are

$$\lambda_1 = \frac{1}{2}\left(2 + \sqrt{-36}\right) = 1 + 3i, \quad \lambda_2 = \frac{1}{2}\left(2 - \sqrt{-36}\right) = 1 - 3i$$

with $p = 1, \;\; q = 3$ so that the general solution is

$$y(x) = e^x\,(A\cos 3x + B\sin 3x).$$

But since

$$y(0) = 4, \qquad y'(0) = 1$$

and

$$y'(x) = A(e^x\cos 3x - 3e^x\sin 3x) + B(e^x\sin 3x + 3e^x\cos 3x),$$

we have

$$y(0) = A = 4, \qquad y'(0) = A + 3B = 1$$

Hence

$$A = 4, \qquad B = -1.$$

Therefore, the particular solution is

$$y(x) = e^{x}(4\cos 3x - \sin 3x). \quad \blacktriangleleft$$

Case III: A Real Double Root

In this case, since $b^2 - 4c = 0,$ then we get only one root

$$\lambda = \frac{-b}{2}$$

hence only one solution

$$y_1 = e^{\frac{-b}{2}x}.$$

To obtain a second independent solution y_2 (needed for a base), we use the reduction method (discussed in Theorem 2, Sec. 2.2). Since one solution is known,

$$y_2 = y_1(x)\int \frac{e^{-bx}}{y_1^2(x)}dx = e^{-\frac{b}{2}x}\int \frac{e^{-bx}}{e^{-bx}}dx = e^{-\frac{b}{2}x}\int dx$$

$$= xe^{-\frac{b}{2}x}.$$

Thus

$$y_1 = e^{\frac{-b}{2}x}, \qquad y_2 = xe^{\frac{-b}{2}x}$$

are linearly independent solutions and therefore are the basis. Hence the corresponding general solution is

$$y(x) = (c_1 + c_2 x)e^{-\frac{b}{2}x}. \quad \blacktriangleleft$$

Example: Solve the following differential equation

a) $y'' + 8y' + 16y = 0$

b) $y'' - 4y' + 4y = 0, \quad y(0) = 3, \quad y'(0) = 1$

Solution:

a) The characteristic equation is

$$\lambda^2 + 8\lambda + 16 = 0$$

and the root is

$$\lambda = \frac{1}{2}(-8 \pm \sqrt{0}) = -4$$

so that the general solution is

$$y(x) = (c_1 + c_2 x)e^{-4x}.$$ ◂

b) The characteristic equation is

$$\lambda^2 - 4\lambda + 4 = 0$$

and the root is

$$\lambda = \frac{1}{2}(4 \pm \sqrt{0}) = 2$$

so that the general solution is

$$y(x) = (c_1 + c_2 x)e^{2x}.$$

But from the initial conditions, $y(0) = 3, \quad y'(0) = 1$ we obtain

$$c_1 = 3, \quad c_2 = -5.$$

Hence the particular solution is

$$y(x) = (3 - 5x)e^{2x}.$$ ◂

Summary of cases *I – III*

Case	*Roots of* $\lambda^2 + b\lambda + c = 0$	*Basis of* $y'' + by + cy = 0$	*General solution of* $y'' + by + cy = 0$
I	Distinct real λ_1 , λ_2	$e^{\lambda_1 x}$, $e^{\lambda_2 x}$	$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
II	Complex conjugate $\lambda_1 = p + iq$ $\lambda_2 = p - iq$	$e^{px}\cos qx$, $e^{px}\sin qx$	$y = e^{px}(A\cos qx + B\sin qx)$
III	Real Double root $\lambda = \frac{-b}{2}$	$e^{\lambda x}$, $xe^{\lambda x}$	$y = (c_1 + c_2 x)e^{\lambda x}$

Activity 2.3

1. Find general solutions for the following differential equations

   a) $y'' - 4y' + 3y = 0$

   b) $y'' - 4y = 0,$

   c) $y'' + 2y' + 10y = 0$

   d) $y'' - 16y' = 0$

   e) $y'' + 4y' = 0$

   f) $y'' - 6y' + 10y = 0$

2. Solve the following initial value problems

   a) $y'' + 2y' + y = 0, \quad y(0) = 1, \quad y'(0) = 0$

   b) $y'' + 14y' + 49y = 0, \quad y(0) = \sqrt{2}, \quad y'(0) = -6\sqrt{2}$

c) $y'' + 2y' + 2y = 0, \quad y(\frac{\pi}{2}) = 0, \quad y'(\frac{\pi}{2}) = -2e^{-\frac{\pi}{2}}$

3. A particle moves on a straight line so that its acceleration is equal to three times its velocity. At $t = 0$ its displacement from the origin is $1\,ft$ and its velocity is $\frac{3}{2}\,ft/\sec$. Find the time when the displacement is $10\,ft$.

4. (*Undamped system, harmonic oscillations*)
   If an iron ball weight $w = 89\,nt$ stretches a spring $10\,cm$, how many cycles per minute will this mass spring system executes? What will the motion be if we pull down the weight an additional $15\,cm$?

5. The three cases of damped motion
   (*Over damping, critical damping and under damping*)
   How does the motion in exercise 4, above, change if the system has damping given by
   a) $c = 200\,kg/\sec$ b) $c = 180\,kg/\sec$ c) $c = 100\,kg/\sec$?
   where c is the constant of damping.

2.4 Euler-Cauchy Equation

Definition:

A differential equation of the form

$$x^2 y'' + bxy' + cy = 0, \tag{1}$$

where b, c are constants, is called the *Euler-Cauchy* equation.

To solve the Euler-Cauchy equation, we let

$$y = x^m \tag{2}$$

Substituting y and its derivatives,

$$y' = mx^{m-1} \quad , \quad y'' = m(m-1)x^{m-2}$$

in to (1), we find

$$x^2 m(m-1)x^{m-2} + bxmx^{m-1} + cx^m$$

$$= x^m (m(m-1) + x^m (bm) + cx^m$$

$$= x^m (m^2 + (b-1)m + c) = 0.$$

For $x \neq 0$, we obtain the characteristic (or auxiliary) equation

$$m^2 + (b-1)m + c = 0 \tag{3}$$

for determining m in (2).

The Three Cases of Solutions

Case I: **Distinct Real Roots.**

If the auxiliary equation (3) has distinct real roots m_1, m_2 we get the basis of solutions

$$y_1 = x^{m_1}, \qquad y_2 = x^{m_2}$$

and a corresponding general solution of (1) is

$$y(x) = c_1 x^{m_1} + c_2 x^{m_2},$$

for all x for which y_1 and y_2 are defined. ◂

Example: Solve the Euler - Cauchy equation

$$x^2 y'' - 4xy' + 6y = 0$$

Solution: The auxiliary equation is

$$m^2 - 5m + 6 = 0$$

The roots are

$$m_1 = 3, \quad m_2 = 2.$$

This gives

$$y_1 = x^3, \; y_2 = x^2,$$

and the general solution is

$$y(x) = c_1 x^3 + c_2 x^2. \qquad \blacktriangleleft$$

Case II: **Complex Conjugate Roots**

If the roots of (3) are complex, they are conjugate, say,

$$m_1 = \mu + iv, \; m_2 = \mu - iv$$

and this gives

$$x^{m_1} = x^{\mu + iv} = x^{\mu} \; x^{iv} = x^{\mu} e^{iv \ln x}$$

$$= x^{\mu} \left[\cos(v \ln x) + i \sin(v \ln x)\right]$$

$$x^{m_2} = x^{\mu - iv} = x^{\mu} \; x^{-iv} = x^{\mu} e^{-iv \ln x}$$

$$= x^{\mu} \left[\cos(v \ln x) - i \sin (v \ln x\right].$$

But Since

$$\frac{1}{2}\left(x^{m_1} + x^{m_2}\right) = x^{\mu}\cos(v\ln x)$$

and

$$\frac{1}{2i}\left(x^{m_1} - x^{m_2}\right) = x^{\mu}\sin(v\ln x),$$

then we obtain the real solutions

$$x^{\mu}\cos(v\ln x) \quad \text{and} \quad x^{\mu}\sin(v\ln x)$$

and the corresponding general solution for all positive x is

$$y(x) = x^{\mu}\left[A\cos(v\ln x) + B\sin(v\ln x)\right]$$ ◂

Example: Solve the differential equation

$$x^2y'' + 7xy' + 13y = 0$$

Solution: The auxiliary equation is

$$m^2 + 6m + 13 = 0.$$

The roots are complex conjugate,

$$m_1 = -3 + \sqrt{9-13} = -3 + 2i,$$

$$m_2 = -3 - \sqrt{9-13} = -3 - 2i,$$

and the corresponding general solution for all positive x is

$$y(x) = x^{-3}\left[A\cos(2\ln x) + B\sin(2\ln x)\right]$$ ◂

Case III: **Double Root**

If (3) has a double root, this root must be

$$\frac{1}{2}(1-b).$$

We then get a first solution

$$y_1(x) = x^{\frac{1}{2}(1-b)}$$

and a second solution y_2 will be obtained by the method of reduction of order (Theorem2, in sec. 2.2). Thus for positive x,

$$y_2 = x^{\frac{1}{2}(1-b)} \int \frac{e^{-b\int \frac{dx}{x}}}{x^{(1-b)}} dx = x^{\frac{1}{2}(1-b)} \int \frac{e^{-b\ln x}}{x^{(1-b)}} dx$$

$$= x^{\frac{1}{2}(1-b)} \int \frac{dx}{x} dx$$

$$= x^{\frac{1}{2}(1-b)} \ln x .$$

That is,

$$y_2 = x^{\frac{1}{2}(1-b)} \ln x,$$

and the corresponding general solution is

$$y(x) = (c_1 + c_2 \ln x) x^{\frac{1}{2}(1-b)}$$ ◂

Example: Solve

$$x^2 y'' - 3xy' + 4y = 0 .$$

Solution: The auxiliary equation $m^2 - 4m + 4 = 0$ has the double root $m = 2$. Hence the basis of real solutions for all positive x are

$$x^2, \qquad x^2 \ln x$$

and the corresponding general solution is

$$y(x) = (c_1 + c_2 \ln x) x^2$$ ◂

Summary of cases *I – III* of *Euler-Cauchy* Equation

Case	*Roots of* $m^2+(b-1)m+c=0$	*Basis of* $x^2y''+bxy'+cy=0$	*General solution of* $x^2y''+bxy'+cy=0$
I	Distinct real m_1, m_2	$y_1=x^{m_1},\ y_2=x^{m_2}$	$y(x)=c_1x^{m_1}+c_2x^{m_2}$
II	Complex conjugate $m_1=\mu+iv,$ $m_2=\mu-iv$	$x^{\mu}\cos(v\ln x),$ $x^{\mu}\sin(v\ln x)$	$y(x)=$ $x^{\mu}\left[A\cos(v\ln x)+B\sin(v\ln x\right.$
III	Real Double root $\frac{1}{2}(1-b)$	$x^{\frac{1}{2}(1-b)},\ x^{\frac{1}{2}(1-b)}\ln x$	$y(x)=(c_1+c_2\ln x)x^{\frac{1}{2}(1-b)}$

Activity 2.4

1. Find a real general solution for the following *Euler-Cauchy* differential equation.

   a) $x^2y''-5xy'+9y=0$

   b) $xy''+2y'=0$

   c) $x^2y''-xy'+2y=0$

   e) $x^2y''-20y=0$

   f) $x^2y''+xy'+y=0$

   g) $x^2y''+7xy'+9y=0$

2. Solve the following *Euler-Cauchy* initial value problems.

   a) $x^2y''+xy'+9y=0,\quad y(1)=2,\quad y'(1)=0$

   b) $x^2y''+3xy'+y=0,\quad y(1)=3,\quad y'(1)=-4$

   c) $x^2y''-2y=0,\quad y(1)=3,\quad y'(1)=1$

3. Show that *Euler-Cauchy* Equations and Constant-Coefficient Equations can be transformed into each other by setting $x=e^t$

2.5 Non Homogeneous Linear Differential Equations

Consider a non homogeneous linear differential equation

$$y'' + f(x)y' + g(x)y = r(x), \qquad (1)$$

where $f(x)$, $g(x)$ and $r(x)$ are continuous functions and $r(x) \neq 0$, and the corresponding homogeneous equation

$$y'' + f(x)y' + g(x)y = 0. \qquad (2)$$

The following Theorem states the relation between (1) and (2).

Theorem 1:

(Relation between solutions of (1) and (2))

a. The difference of two solutions of (1) on some open interval I is a solution of (2) on I.

b. The sum of a solution of (1) on I and a solution of (2) on I is a solution (1) on I.

Proof: Exercise

Definition:

A general solution of the non homogeneous equation (1) on some open interval I is a solution of the form

$$y(x) = y_h(x) + y_p(x), \qquad (3)$$

where

$$y_h(x) = c_1 y_1(x) + c_2 y_2(x)$$

is a general solution of the homogeneous equation (2) on I and $y_p(x)$ is any solution of (1) on I containing no arbitrary constants.

Remark:

1. To solve the non homogeneous equation (1), we have to solve the homogeneous equation (2) and find any particular solution y_p of (1).
2. Any particular solution y_p of (1) must be linearly independent with y_1 and y_2 .
3. To obtain any particular solution $y_p(x)$, there are two methods
    - The Method of *Undetermined Coefficients* and
    - The Method of *Variation of Parameters*

2.5.1 The Method of Undetermined Coefficients

This method applies to differential equations

$$y'' + by' + cy = r(x) , \qquad (*)$$

with constant coefficients and special $r(x)$, namely, *exponential functions*, *polynomials*, *cosines*, *sines*, or *sums or products of such functions*. These $r(x)$ have derivatives of a form similar to $r(x)$ itself. Choose for y_p a form similar to that of $r(x)$ and involving unknown coefficients to be determined by substituting that choice for y_p into $(*)$. The rules of the method are as follows

Rules for the Method of Undetermined Coefficients

(a) **Basic Rule**.

If $r(x)$ in $(*)$ is one of the functions in the first column, in the table given below, choose the corresponding function y_p in the

second column and determine its undetermined coefficients by substituting y_p and its derivatives into (*)

(b) **Modification Rule**.

If a term in your choice for y_p happens to be a solution of the homogeneous equation corresponding to (*), then multiply your choice of y_p by x (*or* $by\,x^2$).

(c) **Sum Rule**.

If $r(x)$ is a sum of functions in several lines of first column, (in the table given below), then choose for y_p the sum of the functions in the corresponding lines of the second column.

Note that we always have to solve the homogeneous equation first.

	Terms in $r(x)$	Choice for y_p
i	ke^{ax}	ce^{ax}
ii	$kx^n \ (n = 0, 1, \ldots)$	$k_n x^n + k_{n-1}x^{n-1} + \ldots + k_1 x + k$
iii	$k\cos bx$	$A\cos bx + B\sin bx$
iv	$k\sin bx$	
v	$ke^{\alpha x}\cos bx$	$e^{\alpha x}(A\cos bx + B\sin bx)$
vi	$ke^{\alpha x}\sin bx$	

Example: Solve the following non homogeneous differential equations by the Method of Undetermined Coefficient.

a) $y'' + 4y = 8x^2$

b) $y'' - 3y' + 2y = 4x + e^{3x}$

c) $y'' - 4y' + 4y = e^{2x}$

Solution:

a) Observe that $y_h(x) = A\cos 2x + B\sin 2x,$ and since $r(x) = 8x^2$ is a polynomial of degree two, then

$$y_p(x) = k_2x^2 + k_1x + k_0.$$

By substituting y_p and y_p'' in the equation we obtain

$$k_2 = 2, \quad k_1 = 0, \qquad k_0 = -1$$

Thus

$$y_p(x) = 2x^2 - 1,$$

and the general solution is

$$\begin{aligned} y(x) &= y_n(x) + y_p(x) \\ &= A\cos 2x + B\sin 2x + 2x^2 - 1. \end{aligned}$$ ◂

b) Observe that

$$y_h(x) = c_1e^{2x} + c_2e^{x},$$

and since $r(x)$ is given as the sum of forms (i) and (ii), then

$$y_p(x) = (ax + b) + ce^{3x}$$

By substituting $y_p, y_p' \text{ and } y_p''$ in the equation we find

$$a = 2, \qquad b = 3, \qquad c = \frac{1}{2}.$$

Hence

$$y_p(x) = 2x + 3 + \frac{1}{2}e^{3x},$$

and the general solution is

$$y(x) = c_1 e^{2x} + c_2 e^{x} + \frac{1}{2}e^{3x} + 2x + 3. \quad \blacktriangleleft$$

c) Since

$$y_h(x) = c_1 e^{2x} + c_2 x e^{2x},$$

and both e^{2x} *and* xe^{2x} are solutions of homogeneous equation, then by modification rule ,

$$y_p(x) = a\,x^2 e^{2x}.$$

By substituting y_p, y_p' and y_p'' in the equation we find $a = \frac{1}{2}$ and hence

$$y_p(x) = \frac{1}{2}x^2 e^{2x}.$$

Thus

$$y(x) = c_1 e^{2x} + c_2 x e^{2x} + \frac{1}{2}x^2 e^{2x}$$

is the general solution is $\blacktriangleleft$

Activity 2.5.1

1. Solve the following non homogeneous differential equations by the Method of Undetermined Coefficients.

a) $y'' - 3y' + 2y = e^x \sin x$

b) $y'' + y = 1 + x + x^2$

c) $y'' - 2y + y = x + e^x$

d) $y'' + y = e^x$

e) $y'' - y = 2e^x + 6e^{2x}$

f) $y'' + 3y' = 28\cosh 4x$

g) $y'' + 8y' + 16y' = 64\cosh 4x$

h) $y'' + 2y' - 35y = 12e^{5x} + 37\sin 5x$

i) $y'' + 6y' + 9y = 50e^{-x}\cos x$

j) $3y'' + 10y' + 3y = 9x + 5\sin x$

2. Solve the following non homogeneous initial value problems by the Method of Undetermined Coefficients.

a) $y'' - y' - 2y = 10\sin x,\ y(0) = 1,\quad y'(0) = -3.$

b) $y'' - 6y' + 13y = 4e^{3x},\qquad y(0) = 2,\qquad y'(0) = 4.$

c) $y'' - 4y' = e^{-2x} - 2x,\qquad y(0) = 0,\qquad y'(0) = 4$

c) $y'' + y' = 2 + 2x + x^2,\qquad y(0) = 8,\qquad y'(0) = -1$

2.5.2 The Method of Variation of Parameters

Consider a differential equation of the form

$$y'' + f(x)y' + g(x)y = r(x)\,, \qquad (1)$$

where f, g and r are continuous functions on some interval I, and a corresponding homogeneous equation

$$y'' + f(x)y' + g(x)y = 0\,. \qquad (2)$$

Recall that the general solution of (2) is given by

$$y_h(x) = c_1 y_1(x) + c_2 y_2(x)\,,$$

where y_1, y_2 form a basis of solutions, and the method of variation of parameters involves in replacing the constants c_1 and c_2 (here regarded as "parameters" in y_h) by functions $u(x)$ and $v(x)$ to be determined so that the resulting function

$$y_p(x) = u(x)y_1(x) + v(x)y_2(x) \qquad (3)$$

is a particular solution of equation (1). By differentiating (3) we obtain

$$y_p' = u'y_1 + uy_1' + v'y_2 + vy_2'\,.$$

To simplify this expression, it is convenient (but not necessary) to set

$$u'y_1 + v'y_2 = 0\,.$$

Then

$$y_p'(x) = uy_1' + vy_2'\,.$$

By differentiating this function we have

$$y_p'' = u'y_1' + uy_1'' + v'y_2' + vy_2''\,.$$

Substituting the expressions for y_p, y_p' and y_p'' into (1) and collecting terms containing u and terms containing v, we obtain

$$u(y_1'' + f(x)y_1' + g(x)y_1) + v(y_2'' + f(x)y_2' + g(x)y_2) + u'y_1' + v'y_2' = r.$$

Since y_1 and y_2 are solutions of the homogeneous equation (3), this reduces to

$$u'y_1' + v'y_2' = r.$$

This gives a second equation relating $u'(x)$ and $v'(x)$, and we have the simultaneous equations

$$\begin{aligned} u'y_1' + v'y_2' &= r, \\ u'y_1 + v'y_2 &= 0. \end{aligned}$$

Multiply the first equation by $-y_2$ and the second by y_2' and add to get

$$u'(y_1y_2' - y_2y_1') = -y_2 r.$$

Thus

$$u' = \frac{-y_2 r}{y_1y_2' - y_2y_1'}.$$

Now multiply the first equation by y_1 and the second by $-y_1'$ and add to get

$$v'(y_1y_2' - y_2y_1') = y_1 r.$$

Thus

$$v' = \frac{y_1 r}{y_1y_2' - y_2y_1'}.$$

Since y_1, y_2 are linearly independent, we have Wronskian of y_1, y_2,

$$W(y_1, y_2) = y_1y_2' - y_2y_1' \neq 0.$$

By integration,

$$u = -\int \frac{y_2 r}{y_1y_2' - y_2y_1'}\,dx, \qquad v = \int \frac{y_1 r}{y_1y_2' - y_2y_1'}\,dx.$$

Substituting them into (3), we obtain

$$y_p(x) = -y_1(x)\int \frac{y_2(x)r(x)}{y_1(x)y_2'(x) - y_1'(x)y_2(x)}\,dx + y_2(x)\int \frac{y_1(x)r(x)}{y_1(x)y_2'(x) - y_1'(x)y_2(x)}\,dx$$

Example: Solve the following non homogeneous differential equations.

a) $y'' - 4y = e^{3x}$

b) $x^2y'' - 4xy' + 6y = 21x^{-4}$

c) $y'' + 16y = 17e^{x}, \qquad y(0) = 6, \qquad y'(0) = -2$

Solution:

a) A basis of solutions of the homogeneous equation on any interval is

$$y_1 = e^{2x}, \qquad y_2 = e^{-2x}.$$

This gives

$$y'_1 = 2e^{2x}, \qquad y'_2 = -2e^{-2x}.$$

and hence

$$y_1y'_2 - y'_1y_2 = -4.$$

The particular solution is

$$y_p(x) = -e^{2x}\int \frac{e^{-2x}\cdot e^{3x}}{(-4)}dx + e^{-2x}\int \frac{e^{2x}\cdot e^{3x}}{(-4)}dx$$

$$= \frac{1}{4}e^{2x}\int e^{x}dx - \frac{1}{4}e^{-2x}\int e^{5x}dx$$

$$= \frac{1}{4}e^{3x} - \frac{1}{20}e^{3x}$$

$$y_p(x) = \frac{1}{5}e^{3x}.$$

But since

$$y_h(x) = c_1e^{2x} + c_2e^{-2x},$$

then the general solution is

$$y(x) = y_h(x) + y_p(x) = c_1e^{2x} + c_2e^{-2}x + \frac{1}{5}e^{3x}. \qquad \blacktriangleleft$$

b) Since

$$y_1 = x^3, \qquad y_2 = x^2$$

are the basis of the solutions of the corresponding homogeneous equation on any interval and

$$y_1' = 3x^2, \qquad y_2 = 2x$$

then

$$y_1 y_2' - y_1' y_2 = -x^4.$$

Re writing

$$x^2 y'' - 4xy' + 6y = 21x^{-4}$$

we get

$$y'' - \frac{4}{x} y' + \frac{6}{x} y = 21x^{-6}.$$

Thus

$$r(x) = 21x^{-6}.$$

Hence the particular solution is

$$y_p(x) = -x^3 \int \frac{(21x^{-6})x^2}{(-x^4)}\,dx + x^2 \int \frac{(21x^{-6})x^3}{(-x^4)}\,dx$$

$$= \frac{1}{2} x^{-4}.$$

But since

$$y_h(x) = c_1 x^3 + c_2 x^2,$$

then the general solution is

$$y(x) = c_1 x^3 + c_2 x^2 + \frac{1}{2} x^{-4}.$$ ◀

c) Since

$$y_1 = \cos 4x,\ y_2 = \sin 4x$$

are the basis of the solutions of the corresponding homogeneous equation on any interval and

$$y_1' = -4\sin 4x, \qquad y_2' = 4\cos 4x,$$

then

$$y_1 y_2' - y_1' y_2 = 4.$$

Thus the particular solution is

$$y_p(x) = -\frac{17}{4}\cos 4x \int e^x \sin 4x dx + \frac{17}{4}\sin 4x \int e^x \cos 4x dx = e^x.$$

But since

$$y_h(x) = A\cos 4x + B\sin 4x,$$

then the general solution is

$$y(x) = A\cos 4x + B\sin 4x + e^x.$$

From the initial condition

$$y(0) = 6, \qquad y'(0) = -2,$$

and

$$y'(x) = -4A\sin 4x + 4B\cos 4x + e^x$$

it follows that

$$y(0) = A + 1 = 6 \quad \Rightarrow \quad A = 5$$

$$y'(0) = 4B + 1 = -2 \quad \Rightarrow \quad B = \frac{-3}{4}.$$

Thus

$$y(x) = 5\cos 4x - \frac{3}{4}\sin 4x + e^x. \qquad \blacktriangleleft$$

Activity 2.5.2

1. Solve the following non homogeneous differential equations by the Method of Variation of Parameters.

a) $y'' + 2y' + y = e^{-x}\cos x$

b) $y'' - 2y' + y = \dfrac{e^x}{x}$

c) $y'' - 2y' + y = e^x \sin x$

d) $y'' - 5y' + 6y = x^3 e^{2x}$

e) $y'' + 2y' + y = e^{-x}\ln x$

f) $y'' - 9y = \sin x$

g) $y'' + y = \sec x$

h) $y'' + y = \sec x$

i) $y'' - y = e^{2x}$

j) $y'' - 2y' + y = e^x$

2. Solve the following non homogeneous Euler- Cauchy differential equations.

a) $xy'' - y' = (3+x)x^2e^x$

b) $4x^2y'' + 8xy' - 3y = 7x^2 0 - 15x^3$

c) $x^2y'' - 4xy' + 6y = 7x^4 \sin x$

d) $x^2y'' + 3xy' - 15y = \dfrac{1}{x}$

e) $x^2y'' + 3xy' + y = 3x^2$

f) $x^2y'' + 3xy' - 15y = x^2e^x$

3. Find the steady-state oscillation of the mass-spring system governed by the equation given below.

a) $y'' + 3y' + 2y = 20\cos 2t$

b) $y'' + 2y' + 5y = -13\sin 3t$

2.6. Systems of Differential Equations

To solve the systems of differential equations, we use the method of solution by elimination. In this method, unknown functions and their derivatives are successively eliminated until one arrives at a single higher order differential equation containing only one unknown function and its derivatives.

Note that here $y := y(t)$, $x := x(t)$.

Now consider systems of the form

$$(a) \qquad x' = a_1 x + b_1 y + f_1$$

$$(b) \qquad y' = a_2 x + b_2 y + f_2,$$

where f_1 and f_2 are given functions and a_1, b_1, a_2, b_2 are constants.

Let us now solve this equation.

$$(a) \Rightarrow \qquad b_1 y = x' - a_1 x - f_1 \qquad (*)$$

Again from (a), we have $x'' = a_1 x' + b_1 y' + f_1'$

$$= a_1 x' + b_1 (a_2 x + b_2 y + f_2) + f_1', \qquad \text{using } (b).$$

$$= a_1 x' + b_1 a_2 x + b_1 b_2 y + b_1 f_2 + f_1'$$

$$= a_1 x' + b_1 a_2 x + b_2 (x' - a_1 x - f_1) + b_1 f_2 + f_1', \qquad \text{using } (*).$$

$$\therefore \quad x'' = (a_1 + b_2) x' + (b_1 a_2 - a_1 b_2) x - b_2 f_1 + b_1 f_2 + f_1'$$

$$\Rightarrow \quad x'' + (-a_1 - b_2) x' + (a_1 b_2 - b_1 a_2) x = b_1 f_2 - b_2 f_1 + f_1'$$

$$\Rightarrow \quad x'' + bx' + cx = r,$$

where $b = -(a_1 + b_2)$, $\quad c = a_1 b_2 - b_1 a_2$, $\quad r(t) = b_1 f_2(t) - b_2 f_1(t) + f_1'(t)$.

By solving this differential equation we get x, then y is obtained from equation $(*)$. ◀

Example: Solve the systems of differential equations:

$$(a) \quad x' = x + y$$
$$(b) \quad y' = x - y + e^t$$

Solution: From (a) we get $y = x' - x$ (*)

Again from (a), we have $x'' = x' + y'$

$= x' + x - y + e^t$, using (b).

$= x' + x - (x' - x) + e^t$, using (*).

$= 2x + e^t$

Thus $x'' - 2x = e^t$

By solving this differential equation we get,

$$x(t) = c_1 e^{\sqrt{2}t} + c_2 e^{-\sqrt{2}t} - e^t \ .$$

Then from (*)we obtain

$$y = x' - x = (\sqrt{2}c_1 e^{\sqrt{2}t} - \sqrt{2}c_2 e^{-\sqrt{2}t} - e^t) - (c_1 e^{\sqrt{2}t} + c_2 e^{-\sqrt{2}t} - e^t)$$

That is,

$$y(t) = c_1(\sqrt{2} - 1)e^{\sqrt{2}t} + c_2(-\sqrt{2} - 1)e^{-\sqrt{2}t}.$$

Hence the general solutions of the systems of differential equations are

$$x(t) = c_1 e^{\sqrt{2}t} + c_2 e^{-\sqrt{2}t} - e^t$$
$$y(t) = c_1(\sqrt{2} - 1)e^{\sqrt{2}t} + c_2(-\sqrt{2} - 1)e^{-\sqrt{2}t}$$ ◀

Activity 2.6

Solve the following systems of differential equations.

a)
$$x' = x + y$$
$$y' = 3x - y + e^t$$

b)
$$x' = 4x - 2y$$
$$y' = 3x - y - 2e^{3t}$$

2.7 Unit Summary

- A differential equation of the form

$$y'' + f(x)y' + g(x)y = r(x), \qquad (1)$$

where f, g and r are continuous functions, is called a linear differential equation of the second order. If $r(x) = 0$, then equation (1) becomes

$$y'' + f(x)y' + g(x)y = 0$$

and is said to be homogeneous differential equation.

If $r(x) \neq 0$, then equation (1) is said to be non homogeneous differential equation.

- A solution of a homogeneous differential equation

$$y'' + f(x)y' + g(x)y = 0,$$

on some interval I, is said to be *general solution* if it expressed by the form

$$y(x) = c_1 y_1(x) + c_2 y_2(x),$$

where y_1 and y_2 are linearly independent solutions.

- Consider a homogenous differential equation

$$y'' + by' + cy = 0 \qquad (1)$$

and its characteristic equation

$$\lambda^2 + b\lambda + c = 0 \qquad (2)$$

Then we have the following

Case	*Roots of (2)*	*Basis of (1)*	*General solution of (1)*
I	Distinct real λ_1 , λ_2	$e^{\lambda_1 x}$, $e^{\lambda_2 x}$	$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
II	Complex conjugate $\lambda_1 = p + iq$ $\lambda_1 = p - iq$	$e^{px}\cos qx$, $e^{px}\sin qx$	$y = e^{px}(A\cos qx + B\sin qx)$
III	Real Double root $\lambda = \frac{-b}{2}$	$e^{\lambda x}$, $xe^{\lambda x}$	$y = (c_1 + c_2 x)e^{\lambda x}$

♦ Consider a homogenous *Euler-Cauchy* differential equation

$$x^2 y'' + bxy' + cy = 0 \qquad (1)$$

and its characteristic (or auxiliary) equation

$$m^2 + (b-1)m + c = 0 \qquad (2)$$

Case	*Roots of (2)*	*Basis of (1)*	*General solution of (1)*
	Distinct real m_1, m_2	$y_1 = x^{m_1}$, $y_2 = x^{m_2}$	$y(x) = c_1 x^{m_1} + c_2 x^{m_2}$
II	**Complex conjugate** $m_1 = \mu + iv$, $m_2 = \mu - iv$	$x^\mu \cos(v\ln x)$, $x^\mu \sin(v\ln x)$	$y(x) =$ $x^\mu\left[A\cos(v\ln x) + B\sin(v\ln x)\right]$
III	**Real Double root** $\frac{1}{2}(1-b)$	$x^{\frac{1}{2}(1-b)}$, $x^{\frac{1}{2}(1-b)}\ln x$	$y(x) = (c_1 + c_2 \ln x)x^{\frac{1}{2}(1-b)}$

♦ A general solution of a linear non homogeneous differential equation

$$y'' + by' + cy = r(x), \qquad (1)$$

is given by

$$y(x) = y_h(x) + y_p(x),$$

where

$$y_h(x) = c_1 y_1(x) + c_2 y_2(x)$$

is the general solution of the corresponding homogeneous equation $y'' + by' + cy = 0$, and $y_p(x)$ is a particular solution of equation (1), given by the formula

$$y_p(x) = -y_1 \int \frac{y_2(x) r(x)}{y_1(x) y_2'(x) - y_1'(x) y_2(x)} dx + y_2 \int \frac{y_1(x) r(x)}{y_1(x) y_2'(x) - y_1'(x) y_2(x)} dx.$$

2.8 Review Problems

1. Solve the following differential equations if one solution y_1 is given

   a) $y'' + \frac{1}{x}y' + \left(1 - \frac{1}{4x^2}\right)y = 0,\ y_1(x) = x^{\frac{1}{2}} \sin x$

   b) $(1 - x^2)y'' - 2xy' + 2y = 0,\ y_1(x) = x$

   e) $xy'' + 2y' + xy = 0, \quad y_1 = \frac{\sin x}{x}$

2. Verify that, for any nonzero constant a,

   $$y_1(x) = e^{-ax}$$

   is a solution of $y'' + 2ay' + a^2 y = 0$. Find the general solution.

3. Solve the following differential equations

   a) $y'' - y' - 2y = \sin 2x$

   b) $y'' + 4y' - 4y = \frac{e^{-2x}}{x^2}; \quad x > 0$

   c) $x^2 y'' + 5xy' + 4y = 0;\ x > 0$

   g) $y'' - 2y' + y = xe^x$

   h) $y'' - 2y' + y = \frac{2e^x}{x^3}$

   i) $xy'' + 4y' = 0, \quad x > 0$

4. Solve the following initial value Problems

   a) $y'' - 9y' + 20y = 0, \quad y(0) = 3, \quad y'(0) = 2$

   b) $y'' - 2y' + y = x^2 - 1,\ y(0) = 2, \quad y'(0) = 1$

   c) $4x^2 y'' + 24xy' + 25y = 0, \quad y(1) = 2, \quad y'(1) = -6$

5. Solve the following non homogeneous differential equations by the Method of Variation of Parameters.

a) $y'' - 4y' + 4y = \dfrac{e^{2x}}{x}$ a) $y'' + y = \csc x + x$

6. Find the steady-state oscillation of the mass-spring system governed by the equation:

$$y'' + 2y' + 4y = \sin 0.2t$$

7. Solve the following systems of differential equations.

a) $x' = 2x - y + \sin t$

$y' = x + 3\cos t,\ x(0) = 1,\ y(0) = -1$

b) $x' + y = \cos t - \sin t$

$y' + x = \cos t + \sin t$